Algorithmic Bias Audit Report

Target File: `./test/enroll_in_advanced_training_.py`

Audit Model: Static Code Bias Detection Model v1.2

Analysis Type: Static Code FATE Analysis

Status: 3 Flags Detected

Executive Summary

This report details instances of algorithmic bias identified within the source code during automated static analysis. The evaluation surfaced exclusionary logic based on protected attributes and/or proxies, which may propagate differential treatment through the Software Development Life Cycle (SDLC).

Detailed Findings

Finding 1: Proxy Bias

Association: POSITIVE

Non-Protected Attribute: years_until_retirement

Biased Code Snippet:

```
if employee["years_until_retirement"] > 0:
```

Issue Description: The code uses 'years_until_retirement' as a proxy for the protected attribute 'Age' to determine eligibility for a positive outcome (advanced training), which may unfairly disadvantage older employees.

Impact Reasoning: Relies on a measurable attribute ('years_until_retirement') as a stand-in for a trait that correlates with a protected group, unfairly impacting groups where that attribute does not accurately indicate the intended outcome.

Finding 2: Threshold/Cutoff Bias

Association: POSITIVE

Non-Protected Attribute: years_until_retirement

Biased Code Snippet:

```
employee["years_until_retirement"] > 0
```

Issue Description: The function uses a fixed numeric threshold on 'years_until_retirement' ('> 0') to determine inclusion in the training program, potentially unfairly excluding employees at or near the retirement age.

Impact Reasoning: Uses a fixed non-protected attribute ('years_until_retirement > 0') with a threshold to determine eligibility or outcomes, where the rigid cutoff may produce biased treatment for individuals near the boundary.

Finding 3: Demographic Bias

Association: POSITIVE

Non-Protected Attribute: years_until_retirement

Biased Code Snippet:

```
if employee["years_until_retirement"] > 0:
```

Issue Description: The code results in differential treatment by granting a positive opportunity (advanced training) based on 'years_until_retirement', which acts as a proxy for age, thereby excluding a specific demographic group (older workers).

Impact Reasoning: Uses a non-protected attribute ('years_until_retirement') to result in differential treatment of individuals based on group membership or proxy grouping.